

Notes: Holmström & Milgrom (Econometrica, 1987)

Aggregation and Linearity in the Provision of Intertemporal Incentives

Contents

1	Big picture	3
2	Model environment and key objects	4
2.1	Single-period baseline environment	4
2.2	Preferences and certainty equivalents	4
2.3	Implementability notation	4
2.4	Why the agent's action space can be rich	5
2.5	T-period extension and enumeration accounts	5
2.6	Brownian limit of frequent small actions	5
2.7	Coarse accounting aggregates	6
3	Core model and theoretical results (all equations + interpretation)	6
3.1	Single-period principal-agent program	6
3.2	Exponential utility separates incentives from participation	6
3.3	Uniqueness when the action space is full dimensional	7
3.4	Dynamic programming decomposition in the T-period model	7
3.5	Optimal multi-period contract: linear in time aggregates	8
3.6	Why the Brownian limit looks static	8
3.7	Brownian implementation formula	9
3.8	Static equivalent and optimal Brownian contract	9
3.9	Coarse monitoring and linearity in profits or other aggregates	10
4	Economic intuition, robustness, and limits	10
4.1	Why Mirrlees-style rare-event penalties fail in a dynamic setting	10
4.2	Why linear contracts generate stable effort pressure	10
4.3	Why richness of the agent's strategy space forces simplicity	11
4.4	When profits alone are enough and when separate accounts matter	11
4.5	What assumptions do the results rely on?	11
5	Worked examples and comparative statics	12
5.1	Quadratic action cost in the one-dimensional Brownian model	12
5.2	Relative performance evaluation	13
5.3	Multiple tasks and separate performance measures	13

6	Short summary	14
7	Conclusion	14

1 Big picture

- **Question.** Why do real-world incentive contracts so often look simple, linear, and based on broad accounting aggregates, even though standard principal-agent theory often predicts highly nonlinear and finely tuned contracts?
- **Setting.** The paper studies an agent who acts repeatedly over time, can react to interim performance information, and is paid at the end of the relationship. Both principal and agent have exponential utility, and the agent's cost of action is monetary.
- **Core challenge.** In one-shot moral-hazard models, optimal contracts can be irregular, difficult to characterize, and extremely sensitive to tail events. The famous Mirrlees-type logic can make near-first-best incentives rely on harsh punishment in rare states. The authors ask whether that remains sensible when the agent can adjust behavior over time after seeing how performance is evolving.
- **Main conceptual move.** Replace the one-shot choice of a distribution with an *intertemporal control problem*. The agent repeatedly chooses actions, observes realized outcomes, and can alter future actions in response. This richer strategy space sharply reduces the usefulness of complicated nonlinear contracts.
- **Main discrete-time result.** In a repeated multinomial model, an optimal compensation scheme depends only on **enumeration aggregates** – the number of times each observable outcome occurs – and it is **linear in those accounts**. The exact sequence of outcomes does not matter.
- **Main continuous-time result.** In a Brownian model where the agent controls the drift of a vector process, the optimal contract is a **linear function of end-of-period account balances**. The dynamic problem is solved as if it were a static normal-linear agency problem.
- **Key aggregation result.** If the principal only observes coarser linear aggregates of the underlying accounts, such as revenues, costs, or profits, then the optimal contract is linear in those coarse aggregates. In particular, if only profits are observed, the optimal contract is linear in profits.
- **Main robustness idea.** Linear rules work well in a rich environment because they deliver the same marginal incentive pressure in ordinary states and do not rely on rare-event punishments. They are much less fragile to model misspecification than highly nonlinear schemes.
- **Why the result is economically interesting.** The paper provides a theory of why actual accounting-based compensation often uses simple performance measures, piece rates, or linear bonus rules rather than the bizarre contracts that static theory can generate.
- **Core takeaway.** Aggregation and linearity are not merely convenient approximations. In dynamic agency problems with repeated action and exponential preferences, they can be exactly optimal.

2 Model environment and key objects

2.1 Single-period baseline environment

- There are states of nature $\theta_0, \dots, \theta_N$.
- The agent chooses an action that is represented in reduced form as a probability vector

$$p = (p_0, \dots, p_N)$$

in a feasible set P inside the probability simplex.

- State θ_i yields two public objects:
 - principal profit π_i ,
 - public outcome x_i on which compensation can be conditioned.
- The principal pays $s(x_i)$ to the agent if outcome x_i is observed.
- The agent's action cost is $c(p; \theta_i)$, which may itself be state dependent.

Interpretation. The paper writes the model in reduced form: the agent is not choosing “effort” directly but rather a distribution over outcomes. This makes the framework flexible enough to capture hidden information, repeated action, project choice, and dynamic strategies.

2.2 Preferences and certainty equivalents

- The principal has exponential utility

$$v(y) = -e^{-Ry},$$

with coefficient of absolute risk aversion R .

- The agent has exponential utility

$$u(y) = -e^{-ry},$$

with coefficient of absolute risk aversion r .

- The agent's reservation utility is summarized by a certain equivalent w .

Interpretation. Exponential utility is crucial. It makes the effect of raising the agent's certainty equivalent a pure level shift, so incentive provision can be separated cleanly from participation.

2.3 Implementability notation

The paper defines three useful objects:

- $S(p, w)$: the set of sharing rules that implement action p with agent certain equivalent w ;

- $P_0(w)$: the set of actions that are implementable at certainty equivalent w ;
- $P^*(w)$: the set of principal-optimal implementable actions at certainty equivalent w .

Interpretation. These sets let the authors separate two questions: first, what actions can be induced at all; second, among those, which action the principal wants.

2.4 Why the agent’s action space can be rich

- In the standard one-shot moral-hazard model, the agent often chooses a one-dimensional effort level.
- Here the authors emphasize richer settings in which the agent acts after observing private signals or keeps acting repeatedly over time.
- A contingent strategy of that sort generally induces a high-dimensional distribution over final outcomes.
- Hence the feasible set P can be much richer than a one-dimensional curve in the simplex.

Interpretation. This “richness” is central. Once the agent has many ways to manipulate outcomes, the principal loses the ability to exploit complicated nonlinear contracts without those contracts being undermined.

2.5 T-period extension and enumeration accounts

- Time is divided into periods $t = 1, \dots, T$.
- In each period the agent chooses an action p^t , possibly as a function of past public history X^{t-1} .
- The principal pays only once, at the end, based on the full realized outcome path X^T .
- Let A_i^T denote the number of times outcome i occurs during the pay period.

Interpretation. The vector $A^T = (A_1^T, \dots, A_M^T)$ is an accounting system: it records how often each observable event occurred. The paper shows that these counts are enough for optimal compensation.

2.6 Brownian limit of frequent small actions

- The continuous-time model treats the agent as controlling the drift of an N -dimensional process $Z(t)$.
- The law of motion is

$$dZ = \mu dt + dB, \tag{1}$$

where B is a driftless Brownian motion with covariance matrix Σ .

- The principal’s end-of-period profits are the sum of the account balances, $\sum_i Z_i(1)$.
- The agent chooses the drift process $\mu(t)$ after observing the public history of Z .

Interpretation. This Brownian model is the continuous-time analog of the repeated multinomial setting when actions are frequent and each period's risk and cost are small relative to risk tolerances.

2.7 Coarse accounting aggregates

- The principal may not observe the full vector $Z(t)$.
- Instead the principal may observe only

$$Y(t) = AZ(t),$$

where A is a $K \times N$ aggregation matrix.

- Examples include observing only total profits, only revenues and costs, or other coarse accounting measures.

Interpretation. This is the formal way the paper talks about accounting aggregation. Different reporting systems correspond to different choices of A .

3 Core model and theoretical results (all equations + interpretation)

3.1 Single-period principal-agent program

The principal's problem in the one-period model is

$$\max_{p,s} \sum_i v(\pi_i - s(x_i))p_i, \quad (2)$$

subject to

$$p \in \arg \max_{p' \in P} \sum_i u(s(x_i) - c(p'; \theta_i))p'_i, \quad (3)$$

and

$$\sum_i u(s(x_i) - c(p; \theta_i))p_i \geq u(w). \quad (4)$$

Interpretation. The first line is the principal's expected utility, the second is the incentive-compatibility constraint, and the third is the agent's participation constraint.

The paper first establishes an existence theorem under compactness and a condition that rules out boundary actions that place zero probability on some public outcomes. The existence result matters because static agency problems can otherwise fail to have an optimum.

3.2 Exponential utility separates incentives from participation

Theorem 2 shows three key facts:

1. $s \in S(p, w)$ if and only if $s - w \in S(p, 0)$;
2. $P_0(w) = P_0$ for all w ;
3. $P^*(w) = P^*$ for all w .

Interpretation.

- Raising the agent's certainty equivalent just shifts the entire wage schedule upward by a constant.
- The set of implementable actions does not depend on the reservation utility.
- The action that is optimal for the principal is independent of the required certainty equivalent.

This is the main algebraic simplification of the paper.

3.3 Uniqueness when the action space is full dimensional

Under Assumption A, the feasible set P has nonempty interior in the N -simplex, costs are deterministic and differentiable, and boundary actions are sufficiently unattractive. Then Theorem 3 says:

For any interior p , the implementing sharing rule is either nonexistent or unique.

Interpretation. In a rich environment, the principal typically does *not* have many distinct contracts that implement the same behavior. That is the opposite of the standard one-dimensional moral-hazard case, where there are usually many incentive schemes consistent with a given action.

Economic message. Once the agent can manipulate the distribution in many directions, the principal's contract choice becomes tightly pinned down. Richer strategic flexibility on the agent side forces simplicity on the principal side.

3.4 Dynamic programming decomposition in the T-period model

Let $w_t(X^t)$ be the agent's continuation-value certain equivalent after history X^t . The agent's dynamic program implies that in each period,

$$p^t(X^{t-1}) \in \arg \max_p \sum_i u(w_t(X^{t-1}, x_i) - c(p; \theta_i)) p_i. \quad (5)$$

From this, Theorem 4 proves that a strategy $\{p^t(X^{t-1})\}$ is implementable if and only if each one-period action is implementable, and the end-of-period sharing rule must decompose as

$$s(X^T) = \sum_{t=1}^T s_t(x^t; p^t(X^{t-1})) + w_0. \quad (6)$$

In account form,

$$s(X^T) = \sum_{t=1}^T s_t(p^t(X^{t-1})) \cdot (A^t - A^{t-1}) + w_0. \quad (7)$$

Interpretation. Dynamic incentive provision is built by stacking one-period implementing rules. Exponential utility is what makes the continuation-value shift additive rather than state dependent in some more complicated way.

3.5 Optimal multi-period contract: linear in time aggregates

Theorem 5 states that an optimal strategy is simply to implement the same single-period optimal action p^* every period. The corresponding optimal compensation rule is

$$s(X^T) = \sum_{t=1}^T s(x^t; p^*) = s(p^*) \cdot A^T. \quad (8)$$

Interpretation.

- The principal does not need to vary incentives over time.
- The principal does not need to respond to the order in which outcomes occur.
- Compensation depends only on final account balances – the counts of different outcomes.

What is aggregated? The detailed path X^T is compressed into the vector of counts A^T . If two performance paths generate the same final count vector, they generate the same pay.

What is not generally true? This does *not* usually mean the optimal contract is linear in total profits. With more than two outcome types, knowing only total profits throws away information that matters for incentives.

3.6 Why the Brownian limit looks static

Before solving the Brownian model, the paper shows in a frequent-action discrete approximation that the agent's single-period certain equivalent is approximately

$$w(\hat{p}) \approx -c(\hat{p}) + \hat{p} \cdot s - \frac{r}{2} \sum_i \hat{p}_i (s_i - \bar{s})^2, \quad (9)$$

where $\bar{s} = \sum_i \hat{p}_i s_i$. The first-order conditions imply an approximate local implementation rule of the form

$$s_j \approx w + c(p) + \frac{r}{2} \sigma_s^2 + c_j(p), \quad (10)$$

where $c_j(p)$ is the relevant marginal-cost term. **Interpretation.** Locally, the contract pays: desired certainty equivalent + reimbursement for effort cost + risk premium + a linear marginal incentive term. This is exactly the structure that becomes exact in continuous time.

3.7 Brownian implementation formula

In the continuous-time model, Theorem 6 shows that if a drift process $\{\mu(t)\}$ is implemented with certainty equivalent w , then the sharing rule must satisfy

$$s(Z^1) = w + \int_0^1 c(\mu(t)) dt \quad (11)$$

$$+ \int_0^1 c'(\mu(t))^T dZ - \int_0^1 c'(\mu(t))^T \mu(t) dt \quad (12)$$

$$+ \frac{r}{2} \int_0^1 c'(\mu(t))^T \Sigma c'(\mu(t)) dt. \quad (13)$$

Interpretation.

- The first integral reimburses the agent for action costs.
- The stochastic integral provides incentives through exposure to realized account innovations.
- The subtracting term removes the expected value of that stochastic incentive term.
- The last term is the risk premium required by the agent.

Economic message. The dynamic contract is linear in increments of the account process. When drift is constant, this collapses to a simple linear function of end-of-period account balances.

3.8 Static equivalent and optimal Brownian contract

The principal's Brownian problem is equivalent to choosing a constant drift μ and agent certainty equivalent w to maximize

$$CE_P(\mu, w) = \mathbf{1}^T \mu - \left(w + c(\mu) + \frac{r}{2} c'(\mu)^T \Sigma c'(\mu) \right) - \frac{R}{2} (\mathbf{1} - c'(\mu))^T \Sigma (\mathbf{1} - c'(\mu)), \quad (14)$$

where $\mathbf{1}$ is the vector of ones.

Theorem 7 then says: if μ^* maximizes this expression, an optimal solution is to instruct the agent to choose constant drift $\mu(t) = \mu^*$ and pay

$$s(Z^1) = c(\mu^*) + c'(\mu^*)^T (Z(1) - \mu^*) + \frac{r}{2} c'(\mu^*)^T \Sigma c'(\mu^*). \quad (15)$$

Interpretation.

- The dynamic agency problem can be solved as if it were static.
- The optimal contract is linear in final account balances.
- The slope on each account is exactly the marginal cost component $c'(\mu^*)$.

Why this is striking. Standard agency models usually need the full nonlinear wage schedule to compute incentives. Here the principal can behave as though linearity had been imposed from the start, and that restriction turns out to be without loss.

3.9 Coarse monitoring and linearity in profits or other aggregates

Suppose the principal does not observe $Z(t)$ directly, only the aggregate process

$$Y(t) = AZ(t),$$

with A a $K \times N$ matrix. Define

$$M(A) = \{\mu \in M : c(\mu) \leq c(\mu') \text{ for all } \mu' \text{ such that } A\mu' = A\mu\}.$$

Theorem 8 says that an optimal solution to the principal's aggregate-monitoring problem is to choose

$$\mu^* \in \arg \max_{\mu \in M(A)} CE_P(\mu, 0), \quad (16)$$

and then use the linear sharing rule implied by the Brownian formula above.

Interpretation.

- If the principal sees only an aggregate, the agent must be induced to choose a cost-minimizing drift among all drifts that generate that same aggregate path.
- The optimal contract remains linear in the observed aggregate.
- If the only observable aggregate is total profits, then the optimal contract is linear in profits.

Accounting implication. The right compensation measure depends on the accounting system. With detailed accounts, the principal may want separate weights on different balances. With only profit information, profit-based pay becomes exactly optimal.

4 Economic intuition, robustness, and limits

4.1 Why Mirrlees-style rare-event penalties fail in a dynamic setting

- In the one-shot normal model, severe punishment in very bad tail states can almost implement the first best.
- But if the agent acts repeatedly and sees interim outcomes, such a contract gives distorted incentives over time.
- The agent works hard only when he is close to the disaster region and relaxes effort otherwise.

Interpretation. A contract built to punish rare bad outcomes is not a good dynamic control device. It creates state-contingent effort in exactly the wrong places.

4.2 Why linear contracts generate stable effort pressure

- A linear rule pays the same marginal reward for performance improvements in ordinary states and extreme states.

- With exponential utility, the agent's continuation incentives are unaffected by wealth effects.
- This leads the agent to choose a constant drift rather than a stop-go effort pattern.

Interpretation. Linearity keeps the incentive system “on” at all times. That is why it is dynamically well behaved.

4.3 Why richness of the agent's strategy space forces simplicity

- If the agent has many contingent actions available, he can exploit finely tuned nonlinear payoffs.
- Complicated contracts rely on assumptions that the agent cannot cheaply alter the probabilities of specific states.
- Once the agent can act after seeing more information, many such assumptions become implausible.

Interpretation. The more flexible the agent is, the less credible it is that the principal can exploit weird local features of the payoff distribution. Robust contracts have to be simple.

4.4 When profits alone are enough and when separate accounts matter

- In the repeated multinomial model, profits alone are generally *not* sufficient unless there are only two outcome types.
- In the multidimensional Brownian model, separate accounts are generally useful because different accounts can deserve different incentive weights.
- However, if the principal only observes profits, or if the agent can only be meaningfully evaluated on profits, the optimal contract is linear in profits.

Interpretation. The paper is not saying that all incentive contracts should use only profit. It is saying that whatever aggregate the accounting system legitimately delivers, the contract should be linear in that aggregate.

4.5 What assumptions do the results rely on?

- **Exponential utility.** This eliminates wealth effects and creates the key shift property.
- **Monetary action costs.** The agent's disutility of effort is represented as a subtraction from final wealth.
- **Stationarity and history independence.** These are used for the strongest aggregation result in the repeated discrete-time model.
- **No private information at contracting.** The model is about moral hazard and hidden information arriving after the contract is signed, not classic adverse selection at the signing stage.

- **Agent controls drift, not covariance.** The Brownian linearity result would generally fail if the agent could also choose the riskiness of the process.

Interpretation. These assumptions are strong, but they are transparent. The paper is valuable because it shows exactly what one must assume to justify linear accounting-based incentives as optimal rather than just convenient.

5 Worked examples and comparative statics

5.1 Quadratic action cost in the one-dimensional Brownian model

Suppose the principal is risk neutral ($R = 0$), the process is one dimensional, and

$$c(\mu) = \frac{k}{2}\mu^2, \quad \Sigma = \sigma^2.$$

Then the principal solves

$$\max_{\mu} \mu - \frac{k}{2}\mu^2 - \frac{r}{2}k^2\mu^2\sigma^2.$$

The optimal drift is

$$\mu^* = \frac{1}{k(1 + rk\sigma^2)}. \quad (17)$$

The corresponding principal payoff is

$$\pi^* = \frac{1}{2k(1 + rk\sigma^2)}. \quad (18)$$

The incentive slope is

$$a^* = c'(\mu^*) = k\mu^* = \frac{1}{1 + rk\sigma^2}. \quad (19)$$

One convenient way to write the contract is

$$s(z) = \frac{k}{2}(\mu^*)^2 + k\mu^*(z - \mu^*) + \frac{r}{2}k^2(\mu^*)^2\sigma^2. \quad (20)$$

Interpretation.

- Higher risk aversion r reduces effort and the slope of pay-for-performance.
- More noise σ^2 also reduces effort and incentive intensity.
- Steeper marginal effort costs (higher k) reduce effort.

Economic message. In this model, the moral-hazard distortion is summarized in one clean wedge, $1 + rk\sigma^2$. The formulas make comparative statics almost trivial.

5.2 Relative performance evaluation

Now suppose the principal also observes a second signal y jointly normal with z , where

$$E[y] = 0, \quad \text{Var}(y) = \delta^2, \quad \text{Cov}(z, y) = \rho\sigma\delta.$$

The optimal sharing rule is linear,

$$s(z, y) = a_1 z + a_2 y + \beta,$$

with

$$a_1^* = \frac{1}{1 + rk\sigma^2(1 - \rho^2)}, \quad a_2^* = -\frac{\sigma}{\delta} \rho a_1^*. \quad (21)$$

Interpretation.

- The extra signal is used only to filter noise out of the performance measure.
- What matters is the conditional variance of z given y , namely $\sigma^2(1 - \rho^2)$.
- If y is perfectly correlated with the noise in z , first best can be approached.

Economic message. The Brownian framework nests relative performance evaluation in a very clean way. Extra information is valuable because it removes risk from the agent's pay, not because it changes the basic linear form of the contract.

5.3 Multiple tasks and separate performance measures

Suppose the agent allocates effort across two activities,

$$z_1 = \mu_1 + \theta_1, \quad z_2 = \mu_2 + \theta_2,$$

with independent shocks and cost

$$c(\mu_1, \mu_2) = \frac{\mu_1^2}{2} + \frac{\mu_2^2}{2}.$$

- If the principal observes only total performance $z = z_1 + z_2$, the optimal contract is linear in z and the agent allocates effort symmetrically.
- If the principal observes z_1 and z_2 separately, the optimal contract is

$$s(z_1, z_2) = a_1 z_1 + a_2 z_2 + \beta,$$

and in general $a_1 \neq a_2$ whenever the two measures have different noise levels.

Interpretation. The less noisy activity gets the steeper incentive weight.

Accounting message. A firm may want to reward cost reduction and revenue creation with different slopes rather than rewarding profits alone. This anticipates the logic of using separate cost centers or revenue centers instead of pure profit centers.

6 Short summary

- The paper asks why optimal incentive contracts are often linear and based on coarse accounting measures.
- Exponential utility makes incentive provision separable from participation.
- In repeated discrete-time settings, optimal compensation is linear in final counts of observable outcomes.
- In the Brownian model, the optimal contract is linear in end-of-period account balances and can be computed from a static certainty-equivalent problem.
- If the principal observes only aggregate accounting data, the optimal contract is linear in those aggregates; if only profits are observed, it is linear in profits.
- The underlying economic reason is robustness: linear rules continue to provide sensible incentives when the agent can react over time and exploit complicated nonlinear schemes.

7 Conclusion

Holmström and Milgrom provide one of the clearest theories of why simple incentive contracts are not merely ad hoc. In their framework, the agent's ability to react over time and the principal's need for a robust contract destroy the appeal of rare-event punishments and highly nonlinear pay schedules. The optimal contract becomes linear in a finite set of accounting balances, and with coarser monitoring it becomes linear in the available aggregates.

The paper therefore gives a strong theoretical foundation for piece rates, linear bonuses, and accounting-based evaluation. It also explains why the right performance measure depends on the firm's information system: detailed accounts justify separate linear weights, while coarser accounting systems justify linear pay in the coarser aggregates.

The broad takeaway is that dynamic agency pushes theory toward the kinds of contracts we actually see. Simplicity is not a defect of optimal incentives in this environment; it is the central result.